

Interchange $\rightarrow$
Row 2 and Row 3

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 8 & 2 & 8 \\ 0 & 0 & 3 & 6 \end{bmatrix} \xrightarrow{\text{Row 1} = \text{Row 1} - \frac{1}{3} \times \text{Row 3}} \begin{bmatrix} 1 & 1 & 0 & 4 \\ 0 & 8 & 2 & 8 \\ 0 & 0 & 3 & 6 \end{bmatrix}$$

Row 2 = Row 2 $\rightarrow$
 $-\frac{2}{3} \times \text{Row 3}$

$$\begin{bmatrix} 1 & 1 & 0 & 4 \\ 0 & 8 & 0 & 4 \\ 0 & 0 & 3 & 6 \end{bmatrix} \xrightarrow{\text{Row 1} = \text{Row 1} - \frac{1}{8} \times \text{Row 2}} \begin{bmatrix} 1 & 0 & 0 & 3.5 \\ 0 & 8 & 0 & 4 \\ 0 & 0 & 3 & 6 \end{bmatrix}$$

$\therefore w_1 = 3.5 \quad w_2 = \frac{4}{8} = \frac{1}{2} \quad w_3 = \frac{6}{3} = 2 \quad \square$

33) Use the vectors $u = (u_1, \dots, u_n)$, $v = (v_1, \dots, v_n)$, and $w = (w_1, \dots, w_n)$ to verify the following algebraic properties of $\mathbb{R}^n$.

a) $(u+v)+w = u+(v+w)$ b) $c(u+v) = cu+cv$ for each scalar c

Solution: a) $u+v = (u_1+v_1, u_2+v_2, \dots, u_n+v_n)$ (Vector addition)

$$(u+v)+w = ((u_1+v_1), (u_2+v_2), \dots, (u_n+v_n)) + (w_1, \dots, w_n)$$

$$= ((u_1+v_1)+w_1, (u_2+v_2)+w_2, \dots, (u_n+v_n)+w_n)$$
 (Vector addition)
$$= (u_1+(v_1+w_1), u_2+(v_2+w_2), \dots, u_n+(v_n+w_n))$$
 (Algebraic property ii)
$$= (u_1, u_2, \dots, u_n) + (v_1+w_1, v_2+w_2, \dots, v_n+w_n)$$
 (Vector addition)
$$= (u_1, u_2, \dots, u_n) + (v_1, v_2, \dots, v_n) + (w_1, w_2, \dots, w_n)$$
 (Vector addition)
$$= u + (v+w) \quad \square$$

b) $c(u+v) = cu+cv$ for each scalar c

$$u+v = (u_1+v_1, u_2+v_2, \dots, u_n+v_n)$$
 (Vector addition)
$$c(u+v) = c(u_1+v_1, u_2+v_2, \dots, u_n+v_n) = (c(u_1+v_1), c(u_2+v_2), \dots, c(u_n+v_n))$$
 (Algebraic property i)
$$(c(u+v)) = (cu_1+cv_1, cu_2+cv_2, \dots, cu_n+cv_n)$$
 (Algebraic property i)
$$= (cu_1, cu_2, \dots, cu_n) + (cv_1, cv_2, \dots, cv_n)$$
 (Vector addition)
$$= c(u_1, u_2, \dots, u_n) + c(v_1, v_2, \dots, v_n)$$
 (Scalar multiplication)
$$= cu + cv \quad \square$$